AI DISASTER RESPONSE COORDINATOR: AN AUTONOMOUS MULTI-AGENT SYSTEM FOR EMERGENCY TRIAGE, RESOURCE ALLOCATION, AND INCIDENT DISPATCH

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***Abstract*—Disaster response operations require rapid interpretation of incomplete incident reports, location-aware resource discovery, capacity protection, responder matching, and coordinated dispatch under severe time constraints. This paper presents the AI Disaster Response Coordinator, an autonomous multi-agent architecture designed for emergency triage, resource allocation, and incident dispatch. The system combines a local Qwen2.5:7B language model with a deterministic fallback path, agentic retrieval-augmented generation over Tamil Nadu disaster-management procedures, specialized shelter, hospital, and volunteer agents, and a planning agent that synthesizes operational constraints. A human-in-the-loop approval boundary separates recommendation generation from irreversible dispatch actions. Geographic reasoning uses OpenStreetMap data and the Open Source Routing Machine, while state and transactional updates are maintained through a persistent database layer. The prototype models 380 verified relief shelters, 210 healthcare facilities, and 305 certified responders. Evaluation covers triage reliability, shelter district-priority matching, capacity protection, volunteer skill matching, latency, offline behaviour, and transactional safety. Across the isolated test harness, the implementation achieved 100% assertion-pass rate, zero shelter overbooking, correct atomic capacity release, and robust fallback behavior under simulated local-model timeout and offline conditions. The results indicate that a modular agent architecture with deterministic safeguards can provide a practical foundation for explainable and governable emergency coordination.**

***Index Terms*—disaster response, multi-agent systems, emergency triage, agentic RAG, resource allocation, human-in-the-loop, OpenStreetMap, OSRM, Qwen2.5, deterministic fallback.**

NOMENCLATURE

AI	Artificial Intelligence
DEOC	District Emergency Operations Center
ETA	Estimated Time of Arrival
GIS	Geographic Information System
LLM	Large Language Model
MAS	Multi-Agent System
NDMA	National Disaster Management Authority
OSM	OpenStreetMap
OSRM	Open Source Routing Machine
RAG	Retrieval-Augmented Generation
TN-SDMA	Tamil Nadu State Disaster Management Authority

I. INTRODUCTION

High-impact disasters create an information and coordination problem in which incident descriptions, locations, available resources, capacity limits, responder skills, and operating procedures must be combined quickly. The project report frames this as a coordination "trilemma": response coordination efficiency, resource precision, and operational safety must be satisfied together. Traditional emergency communication relies on fragmented channels, manual data entry, and human dispatch decisions. During a rapidly evolving event, these practices can create shelter overbooking, delayed medical routing, unassigned specialized responders, and incomplete notifications.

The proposed coordinator treats emergency response as a bounded autonomous workflow rather than a single monolithic language-model prompt. A distress call is first converted into a structured incident representation. An assessment agent assigns a severity and objective set, after which specialized agents

discover shelters, hospitals, volunteers, and applicable operating procedures. A planning agent synthesizes the retrieved information into a candidate response plan. The plan then crosses a formal human approval boundary before any external dispatch or capacity mutation is executed.

The architecture is motivated by recent work on large language models in disaster management, which identifies detection, tracking, analysis, and action as important operational roles for language-model systems [1], [2]. Multi-agent systems provide a natural abstraction for heterogeneous emergency actors and distributed coordination [3]. Retrieval grounding is used to connect model reasoning to operational manuals and structured records rather than relying only on parametric model knowledge [4].

The principal contributions are: (1) a modular multi-agent emergency workflow with explicit specialized roles; (2) a local LLM plus deterministic fallback for bounded reasoning; (3) agentic RAG with semantic chunking and sufficiency evaluation; (4) district-aware shelter allocation, atomic capacity protection, and skill-constrained volunteer matching; (5) a human-in-the-loop governance boundary; and (6) an isolated evaluation harness covering functional, reliability, latency, and transactional criteria.

II. RELATED WORK

Artificial intelligence has long been applied to disaster prediction, situational awareness, evacuation planning, and resource allocation. More recent work has shifted toward LLM-based disaster information processing. Lei et al. survey LLM applications across disaster phases and emphasize the need to connect language-model capabilities with operational response tasks [1]. A 2025 interdisciplinary review similarly groups LLM applications around detection, tracking, analysis, and action, while noting challenges in deployment and information management [2]. These findings motivate a system that does not stop at textual summarization but connects assessment to constrained resource actions.

Multi-agent systems are relevant because emergency management contains heterogeneous participants with different objectives and capabilities. Recent agentic vision-language work demonstrates how specialized agents can support post-disaster assessment, emergency alerts, resource allocation, and recovery planning [3]. The present architecture adopts specialized agents for assessment, retrieval, shelter discovery, volunteer allocation, hospital discovery, planning, reflection, and communication. Frameworks such as AutoGen illustrate how multiple LLM-enabled agents can coordinate with human inputs and tools [14]. Knowledge-graph-based LLM decision support has also been explored for evidence-based emergency decision-making [15].

The retrieval component follows the general RAG principle of combining parametric model knowledge with non-parametric external memory [4]. ReAct further demonstrates how language-model reasoning can be interleaved with actions that query external sources or environments [5]. Sentence-BERT provides an efficient basis for semantic similarity search through sentence embeddings and cosine similarity [6]. These ideas are combined in the project as an agentic retrieval pipeline that retrieves operational guidance and evaluates whether the retrieved context is sufficient before planning.

The local model path is based on the Qwen family. Qwen2.5 is a modern instruction-tuned model family designed for general language tasks and reasoning [7]. Local inference reduces dependence on remote services and allows incident information to remain within the deployment boundary. However, LLM generation can hallucinate plausible but unsupported content; surveys of hallucination research therefore motivate retrieval grounding, structured outputs, and deterministic validation [8].

Geospatial response requires both a map data source and a routing engine. OpenStreetMap provides collaboratively maintained, reusable geographic data [9], while OSRM supports real-time routing over OpenStreetMap data [10]. The project uses these services to translate incident coordinates and resource locations into distance and ETA information for operational decisions. Transformer-based language models provide the underlying attention mechanism used by modern LLMs [11].

Finally, disaster-response governance cannot be reduced to model accuracy alone. The Sendai Framework emphasizes disaster-risk understanding, governance, investment in resilience, and preparedness for effective response [12]. The National Incident Management System provides a structured basis for incident coordination and interoperable response [13]. The proposed human approval boundary aligns with these governance-oriented principles by ensuring that an autonomous recommendation is not equivalent to an autonomous authorization.

III. PROBLEM STATEMENT AND OPERATIONAL CONTEXT

The problem addressed is how an autonomous multi-agent architecture can achieve sub-minute emergency response planning while maintaining spatial accuracy, zero shelter overbooking, resource commitments, and deterministic human governance. A distress report may contain only a reporter identity, location, number of people affected, and free-form description. The system must infer the operationally relevant fields, classify severity, identify applicable procedures, locate suitable resources, check constraints, and produce a dispatch-ready plan.

The project models the Tamil Nadu State Disaster Management Authority and district-level District Emergency Operations Center topology. The operational data layer contains 38 administrative districts, verified relief shelters, healthcare facilities, and a statewide volunteer registry. This context is important because district-aware allocation can be preferable to simply selecting the geographically nearest resource: local routing, district priorities, capacity, and operational suitability can all affect the final decision.

The cost of coordination failure is operational rather than merely computational. Shelter overbooking can redirect evacuees toward unsafe or distant facilities. Mismatched responders can delay specialized rescue. A hospital recommendation

without adequate capacity can create downstream overload. Missing or malformed notifications can prevent responders and citizens from receiving the required operational instruction. Therefore, the system explicitly measures capacity integrity, skill matching, dispatch correctness, and failure handling in addition to language-model output quality.

IV. PROPOSED AI SYSTEM

A. System Architecture

The end-to-end architecture is organized as a sequence of bounded processing stages. Distress Call Intake receives the incident narrative, location, affected population, and description. The Assessment Agent converts the narrative into a structured severity and objective representation. Agentic RAG retrieves applicable disaster-management procedures. Shelter Finder, Volunteer Agent, and Hospital Finder independently evaluate their respective resource domains. The Planning Agent synthesizes the candidate resources and identifies gaps or conflicts. The resulting plan enters the human-in-the-loop approval boundary. A rejection activates the Reflection Agent for replanning; an approval enables Atomic Commitment and Dispatch followed by incident closure and capacity release.

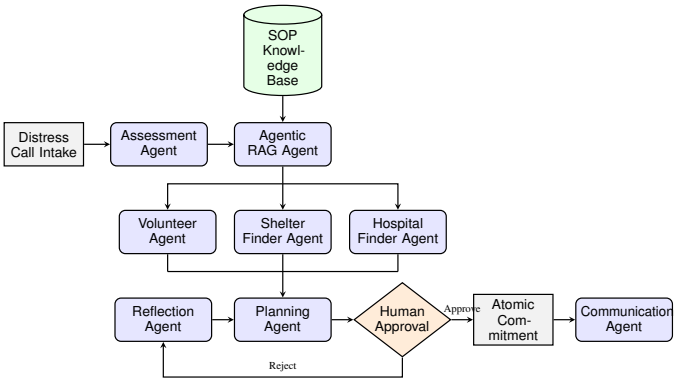


Fig. 1. End-to-end multi-agent coordination architecture

B. Agent Roles and Capabilities

Assessment Agent: analyzes the raw incident narrative, determines triage priority, calculates an objective severity index, and produces mandatory emergency resource categories. The project implementation uses a local Qwen2.5:7B model with a 30-second bounded execution path and a deterministic fallback when the local model is unavailable or unresponsive.

Agentic RAG Agent: retrieves operational guidance from the disaster SOP knowledge base. The knowledge base contains cyclone response, flood response, earthquake response, emergency medical first response, evacuation and shelter safety, and volunteer responder safety material. The retrieval layer uses semantic chunking, Sentence-BERT embeddings, an in-memory vector store, and a sufficiency evaluator.

Shelter Finder Agent: searches the 380-shelter registry and applies district preference, capacity, occupancy, and location constraints. It is responsible for preventing capacity violations rather than simply ranking by distance.

Volunteer Allocation Agent: matches incident requirements against 305 certified responders using a multi-criteria skill matrix. Required capabilities include flood rescue, water rescue, boat operations, first aid, medical assistance, evacuation and location handling, search and rescue, shelter management, and logistics and food distribution.

Hospital Finder Agent: evaluates 210 healthcare facilities using spatial proximity, service suitability, and trauma-related considerations. The agent returns ranked facilities and supports downstream planning.

Planning Agent: combines the outputs of the specialized agents into a coherent rescue plan. It identifies resource gaps, resolves conflicts, and prepares the candidate plan for human review.

Reflection Agent: operates only after rejection or failed validation. It consumes human feedback, updates constraints, and generates a revised plan instead of allowing the system to continue with an unapproved action.

Communication Agent: converts approved operational actions into target-specific messages for responders, hospitals, shelters, volunteers, and affected citizens. The report specifies structured multi-target messages and email/OSRM-linked operational dispatch pathways.

C. Human-in-the-Loop Governance

The approval boundary is a central safety mechanism. The system may autonomously analyze, retrieve, rank, and propose, but an incident commander must explicitly approve the candidate plan before atomic capacity commitment or external dispatch. This design avoids equating model confidence with operational authorization. Approval and rejection are represented as explicit states, and rejected plans are routed to reflection rather than silently executed.

V. DATASETS AND KNOWLEDGE SOURCES

The project report defines four primary data sources. The Relief Center and Shelter Dataset contains 380 verified facilities across 38 administrative districts with fields such as name, district, address, latitude, longitude, capacity, occupancy, contact information, facility type, and operational status. The Hospital Network Dataset contains 210 facilities with hospital type and service attributes. The Skilled Volunteer Registry contains 305 certified responders, including contact information, skill classifications, and availability state. The SOP knowledge base stores operational procedures as structured Markdown documents.

The knowledge base includes cyclone response, flood response, earthquake response, emergency medical first response, evacuation and shelter safety, and volunteer responder safety. The retrieval pipeline therefore grounds recommendations in operational documents rather than asking the model to reproduce procedures from memory. The report also specifies regulatory and district-level context as part of the retrieval corpus.

TABLE I
OPERATIONAL DATASETS AND THEIR ROLES.

Dataset	Records	Operational role
Relief shelters	380	District-aware shelter discovery, capacity protection
Hospitals	210	Proximity and service-based medical ranking
Volunteers	305	Skill matching and availability assignment
SOP KB	6 domains	Retrieval grounding for emergency procedures

VI. AGENTIC RAG IMPLEMENTATION

The RAG subsystem follows a structured retrieval process. Documents are normalized and segmented into semantic chunks rather than arbitrary character boundaries. Sentence-BERT embeddings convert chunks into vector representations, and an in-memory vector store supports similarity search. Given an incident query, the Agentic RAG Agent retrieves the highest-scoring passages and evaluates whether the evidence is sufficiently specific for the current incident.

The project defines a dynamic sufficiency evaluator with a threshold of 0.65. When the top retrieved context is insufficient, the agent expands or rewrites the query and retrieves again. This is important for emergency SOPs because a generic phrase such as "water rising" may not uniquely determine the appropriate response. Query expansion can incorporate incident type, location, affected population, and operational constraints so that the retrieved passages become more actionable.

The isolated retrieval evaluation reported precision@3 of 0.92, recall@5 of 0.96, and a sufficiency evaluation accuracy of 94%. These measurements indicate that the retrieval subsystem is capable of surfacing relevant procedural material, while the sufficiency gate provides a second control against passing weak context to the planning stage.

VII. WORKFLOW, STATE MANAGEMENT, AND COORDINATION

The coordinator is implemented as a deterministic directed workflow governed by a state machine. The principal states are incident ingestion and geocoding, multi-factor assessment, agentic RAG retrieval, parallel asset discovery, planning and gap identification, human approval, reflection or atomic reservation and dispatch, and finally mission completion and shelter release. Explicit state transitions make the lifecycle inspectable and support rollback of transactional operations.

Short-term contextual memory contains incident identifiers, normalized incident data, retrieved context, agent outputs, and the current plan. Long-term persistent memory stores incident history, mission status, shelter occupancy and capacity, volunteer profiles and assignments, and state transitions. The project uses SQLite/PostgreSQL-compatible persistence with transactional updates and atomic rollback semantics for capacity changes.

Inter-agent communication uses structured, schema-validated dictionaries rather than unrestricted natural-language hand-offs. The use of Pydantic-style validation or equivalent schemas prevents malformed outputs from cascading between agents. This is especially important when an upstream model output contains a valid-looking sentence but misses a required field such as district, capacity, skill, or approval state.

Concurrency is handled through explicit coordination mechanisms. The report describes a scenario in which multiple incidents might request the same shelter simultaneously. An atomic capacity reservation checks occupancy plus requested load before committing the update. If the condition fails, the transaction aborts and the mission remains pending rather than creating an overbooking.

VIII. GEOSPATIAL AND OPERATIONAL SERVICES

The geospatial layer uses OpenStreetMap and OSRM. OpenStreetMap supplies geographic objects and coordinates, while OSRM supplies route distance and estimated travel time. The system can therefore compare resource candidates using more than textual proximity. Geocoding converts user-provided location descriptions into coordinates, after which shelter, hospital, and responder locations can be evaluated relative to the incident.

The project specifically uses district-aware proximity and capacity logic for shelter selection. A candidate may be geographically close but unsuitable because it belongs to the wrong district, lacks capacity, or has an unfavorable operational status. The combination of geographic routing and deterministic constraints is therefore more appropriate than asking an LLM to estimate distances or capacities from narrative context.

IX. IMPLEMENTATION ARCHITECTURE

The prototype is organized as a Python-based service with FastAPI providing REST endpoints and a Streamlit interface supporting operational visualization. The application layer coordinates the agent workflow, state machine, notifications, and mission lifecycle. The data layer stores incident, shelter, hospital, volunteer, and mission state. External services include OpenStreetMap, OSRM, geocoding, and email/notification delivery. Local LLM inference is handled through Ollama with Qwen2.5:7B.

The implementation is intentionally hybrid. The LLM performs semantic interpretation and explanation where language understanding is useful; deterministic functions perform spatial calculations, capacity checks, database transactions, and bounded fallback decisions. This division reduces the amount of operational authority assigned to probabilistic generation. It also allows the same workflow to continue when the local model times out or an external service is unavailable.

X. EXPERIMENTAL EVALUATION

Evaluation was conducted with an isolated temporary database and a deterministic test harness. The report defines four broad performance categories: allocation and matching precision, computational latency, reliability and offline resilience, and communication/dispatch fidelity. The tests target concrete invariants rather than subjective model quality alone.

TABLE II
SUMMARY OF REPORTED EVALUATION RESULTS.

Metric	Observed result
Automated assertions	45/45 passed (100%)
Shelter overbooking	0 violations
Atomic capacity	50/50 successful scenarios
Capacity release	50/50 correct releases
Volunteer matching	Correct exclusion of unqualified skills
RAG precision@3	0.92
RAG recall@5	0.96
RAG sufficiency	94%
Warm-path latency	12.5–14.1 s

The triage and inference tests compare local Qwen2.5:7B execution with a bounded deterministic fallback. The fallback is exercised when a simulated timeout or offline condition is introduced. The evaluation reports that the fallback preserves the required structured response behavior rather than allowing the mission workflow to fail silently.

Spatial allocation tests verify district-priority behavior, capacity protection, atomic approval, atomic release, and double-release immunity. The latter is important because an lifecycle may encounter repeated completion or retry messages; a second release must not decrement capacity twice. Volunteer tests verify that required skills are honored and that an unavailable or unqualified responder is not assigned merely because they are geographically close.

XI. COMPARISON AND DISCUSSION

TABLE III
COMPARISON OF COORDINATION CAPABILITIES.

Capability	Conventional	Proposed system
Dispatch plan	Manual lookup; slow	Multi-agent synthesis
Shelter assign	Spreadsheet selection	District + capacity logic
Overbooking	Human error possible	Atomic reservation
Volunteer	Ad-hoc lookup	Skill-matrix matching
Offline mode	Paper-based	Deterministic fallback
Governance	Manual	Formal approval boundary
Notification	Manual calls/SMS	Multi-target structured

The principal advantage is not that the LLM replaces emergency personnel. Instead, the system decomposes the coordination problem so that language understanding, retrieval, geospatial computation, capacity accounting, and human authorization have separate responsibilities. This separation improves testability and makes failure modes more explicit. It also allows individual agents to be replaced or upgraded without redesigning the complete workflow.

The project nevertheless has important limitations. The evaluation uses an isolated test harness and project-defined datasets rather than a live emergency deployment. Routing and geocoding depend on external geographic services. A local 7B model can still produce incorrect semantic interpretations, and deterministic fallback cannot replace domain expertise for ambiguous incidents. The system therefore remains a decision-support and coordination prototype rather than a certified emergency command system.

XII. SCALABILITY, SAFETY, AND FUTURE WORK

The architecture is horizontally scalable because the FastAPI service is stateless with respect to long-term mission state, which is persisted separately. Additional agent workers can be introduced for parallel discovery while the database remains the source of truth for resource state. The project also describes migration from SQLite toward PostgreSQL/PostGIS for larger deployments and improved geospatial querying.

Future directions include dynamic flood-indundation or weather-map integration, satellite-radar interpretation, multimodal citizen reporting, automated two-way SMS or WhatsApp dispatch, and decentralized low-bandwidth communication. These extensions should preserve the same safety principle: probabilistic perception may generate candidate information, but high-impact commitments should remain constrained by deterministic validation and explicit human governance.

A field deployment would additionally require larger-scale validation with emergency-management personnel, failure-injection testing against real service outages, security review of API credentials and operational endpoints, and formal assessment of fairness and accessibility. The project report also highlights the need to verify geocoding assumptions and routing behavior in locations where roads or infrastructure may be missing or temporarily inaccessible.

XIII. PROTOTYPE INTERFACE AND OUTPUTS

The prototype provides a Streamlit-based operational interface for incident intake, AI assessment, resource recommendation, geospatial allocation, and human approval. The principal outputs are arranged below as figure panels, showing the operational path from incident submission and AI assessment to resource recommendation and human approval.



Fig. 2. AI Disaster Response Coordinator dashboard and navigation.

The outputs demonstrate the operational path from incident submission and AI assessment to resource recommendation, geospatial shelter allocation, explanation, and explicit human approval.

XIV. CONCLUSION

This paper presented an AI Disaster Response Coordinator built as a modular autonomous multi-agent system for

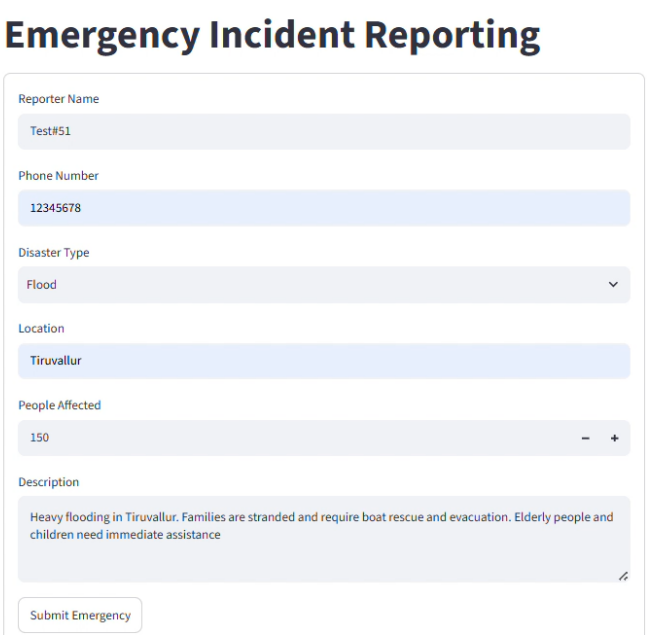


Fig. 3. Emergency incident reporting interface.

emergency triage, resource allocation, and incident dispatch. The design combines local LLM reasoning, agentic RAG, specialized resource agents, geospatial routing, deterministic capacity and skill constraints, persistent state, and a human-in-the-loop approval boundary. The prototype operates over 380 shelters, 210 hospitals, and 305 certified responders and was evaluated with functional, transactional, retrieval, latency, and resilience tests.

The evaluation demonstrates that the architecture can maintain operational invariants such as zero shelter overbooking, correct capacity release, and skill-constrained volunteer assignment while providing a bounded language-model path for incident interpretation. The broader lesson is that emergency AI should be designed as a governed system of cooperating components rather than as an unconstrained conversational agent. Combining semantic flexibility with deterministic safeguards provides a stronger basis for reliable disaster-response coordination.

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